

Aditya Kumar

Backend-Heavy Full Stack Developer | AI Integration & Agentic Systems | System Design
+91-7004271114 • adiakumar27@gmail.com • linkedin.com/in/adityakumar70042 • github.com/adityakumar565

PROFESSIONAL SUMMARY

Backend-heavy Full Stack Developer with **2 years of Java/Spring Boot experience** architecting scalable, distributed systems. Engineered **15+ production APIs** and optimized 30+ legacy endpoints, **slashing latency by 80%** through systems-level performance engineering (memory pre-allocation, connection sorting, Data-Oriented Design). Specializes in integrating **AI agents and RAG pipelines** directly into backend service architecture: architected **Project Nexus**, a polyglot microservice platform binding a Java/Spring Boot state engine to a Python/FastAPI AI inference layer with **LangChain4j and Spring AI agentic workflows**.

TECHNICAL SKILLS

Backend & System Design: Java, Spring Boot, RESTful API Design, Microservices Architecture, Distributed Systems, Data-Oriented Design (DOD)

AI / Agentic Systems: LangChain4j, Spring AI, Retrieval-Augmented Generation (RAG), Vector Databases, AI Agent Integration, Python, FastAPI

Languages & Testing: Java, Python, SQL, C++, JavaScript (ES6+), JUnit, Mockito

Cloud & DevOps: Docker, AWS (EC2/S3), CI/CD Pipelines, PostgreSQL, Git, GitHub

Frontend Integration: React.js, HTML5/CSS3, RESTful Client Binding

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)
Software Engineer

Kolkata, India
Jun 2024 – Present

- Spearheaded **system design** and cross-functional architecture decisions, leading **CI/CD integrations** that coordinated multi-team deployment of **15+ production microservices**.
- Slashed API latency by 80%** on data-heavy payloads (**2,000+ branched points**) by engineering custom pagination, pre-allocating memory, and optimizing database connection sorting — a backend performance overhaul at the systems level.
- Architected **RESTful backend interfaces for AI agent / chatbot integration**, designing the data collection and preprocessing pipeline that structured **10,000+ daily telemetry events** for downstream ML training.

PROJECTS

Project Nexus (Agentic Workflow & RAG Platform)

Jan 2026 – Present

Java, Spring Boot, React.js, Python, FastAPI, LangChain4j

- Architected a **polyglot microservice ecosystem**, binding a React.js frontend to a **Java/Spring Boot state engine**. Engineered **PostgreSQL recursive CTEs** to traverse graph trees (tested up to 50,000+ mock nodes) and **eliminate N+1 query bottlenecks** — a core system design challenge.
- Built a **Python/FastAPI service layer** to isolate AI inference from the core backend, implementing **agentic Retrieval-Augmented Generation (RAG) pipelines** via **LangChain4j and Spring AI** to stream thousands of structured JSON outputs into the database.
- Profiled execution hot loops to maximize hardware performance, applying **Data-Oriented Design (DOD)** principles like flattened 1D arrays and monomorphic inline caching to prevent CPU L1 cache pollution, achieving **sub-15ms execution times**.

EDUCATION

Institute of Engineering and Management (IEM)
Bachelor of Technology (B.Tech) • CGPA: 9.0

Kolkata, India
Aug 2020 – May 2024